```
import pandas as pd
from matplotlib import pyplot as plt
%matplotlib inline
```

```
df=pd.read_excel('/content/skin_logistic_regression_data.xlsx')
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(df[['Acne_Count']],df.Acne_Severity,train_size=0.70)
```

```
X_test
```

	Acne_Count
3	18
25	0
161	11
93	7
7	9
50	4
99	10
116	18
113	4
164	0
157	13
187	0
178	9
141	7
82	5

Variables Terminal

Commands + Code + Text Run all

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
```

```
X_test
```

141	7
82	5
2	17
20	14
170	13
175	13
181	7
108	10
115	3
38	6
179	17
150	14
100	3
183	7
158	10
168	8
117	19

```
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

```
y_predicted=model.predict(X_test)
```

```
y_predicted
```

```
array([1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1, 0,
       0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1,
       1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1])
```

```
model.predict_proba(X_test)
```

```
       [0.45954559, 0.54045441],
       [0.62051608, 0.37948392],
       [0.75871561, 0.24128439],
       [0.05853469, 0.94146531],
       [0.14222265, 0.85777735],
       [0.18694337, 0.81305663],
       [0.18694337, 0.81305663],
       [0.62051608, 0.37948392],
       [0.38009904, 0.61990096],
       [0.85809563, 0.14190437],
       [0.69395926, 0.30604074],
       [0.05853469, 0.94146531],
       [0.14222265, 0.85777735],
       [0.85809563, 0.14190437],
       [0.62051608, 0.37948392],
```

```
       [0.05853469, 0.94146531],
       [0.14222265, 0.85777735],
       [0.54110293, 0.45889707],
       [0.94160906, 0.05839094],
       [0.14222265, 0.85777735],
       [0.75871561, 0.24128439],
       [0.89345399, 0.10654601],
       [0.81345323, 0.18654677],
       [0.07937552, 0.92062448],
       [0.89345399, 0.10654601],
       [0.24176278, 0.75823722]])
```

```
model.score(X_test,y_test)
```

```
0.7666666666666667
```

```
model.coef_
```

```
array([[0.32695572]])
```

```
model.intercept_
```

```
array([-2.78042936])
```

```
import math
def sigmoid(x):
  return 1 / (1 + math.exp(-x))
```

```
def prediction_function(acne_count):
    z = 0.32695572 * acne_count -2.78042936 # 0.04150133 ~ 0.042 and -1.52726963 ~ -1.53
    y = sigmoid(z)
    return y
```

```
acne_count=1
prediction_function(acne_count)
```

```
0.07918489941518299
```